

Ruihan Henry Liu

PH.D. CANDIDATE · UNIVERSITY OF CALIFORNIA, BERKELEY

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Research Interests

I work at the intersection of observational and computational cosmology. My research focuses on the large-scale structure of the Universe and the cosmic microwave background, with an emphasis on probing baryonic matter and feedback through the Sunyaev-Zel'dovich effects and weak gravitational lensing. I make use of data from cosmological surveys such as **ACT, DESI, and HSC**, together with hydrodynamical simulations including **IllustrisTNG, SIMBA, and CAMELS**. I also develop and apply Bayesian inference and machine learning methods for cosmological data analysis.

Education

University of California, Berkeley

Berkeley, California

DOCTOR OF PHILOSOPHY (PH.D.) IN PHYSICS

Expected May 2027

Advisors: Uroš Seljak and Simone Ferraro

MASTER OF ARTS (MA) IN PHYSICS

Dec. 2024

University of British Columbia

Vancouver, B.C., Canada

B.Sc. (HONOURS) IN PHYSICS AND MATHEMATICS

May 2021

Thesis: *On Bayesian Baseline Removal and Signal Extraction Methods in Astronomy*

Advisor: Douglas Scott

Research Experience

University of California, Berkeley

Berkeley, California

GRADUATE RESEARCHER

Sept. 2021 - Present

- Cosmology: Large Scale Structure and the Cosmic Microwave Background. Focusing on probes of baryonic matter through the Sunyaev-Zel'dovich effects and weak lensing, as well as computational and statistical approaches to cosmology.

University of California, Berkeley · VISITING STUDENT RESEARCHER (ADVISOR: U. SELJAK)

May 2020 - Aug. 2020

Canadian Institute for Theoretical Astrophysics · UNDERGRADUATE RESEARCHER (ADVISOR: P. C. BREYSSE)

May 2019 - Aug. 2019

University of British Columbia · NSERC USRA UNDERGRADUATE RESEARCHER (ADVISOR: D. SCOTT)

May 2018 - Jul. 2019

Imperial College London · IROP EXCHANGE RESEARCH STUDENT (ADVISOR: R. TROTTA)

Jul. 2018 - Aug. 2018

NRC Herzberg Institute of Astrophysics · CO-OP MACHINE LEARNING STUDENT (ADVISORS: S. FABBRO AND A.

McCONNACHIE)

Jan. 2018 - Apr. 2018

Publications and Preprints

First Author Publications:

- [1] Liu, R. H., Seljak, U., Ferraro, S., Hadzhiyska, B., *et al.*, **Model-Independent Measurement of Baryon Gas Fractions through Galaxy-Galaxy Lensing and the Kinematic Sunyaev-Zel'dovich Effect**, *in preparation, to be submitted to Phys. Rev. D*.
- [2] Liu, R. H., Ferraro, S., Hadzhiyska, B., **Baryons Beyond Halos: Unbound Gas, Stars, and the Sunyaev-Zel'dovich Effect**, *in preparation, to be submitted to Phys. Rev. D*.
- [3] Liu, R. H., Hadzhiyska, B., Ferraro, S., Bose, S., Hernández-Aguayo, C., **Fast Baryonic Field Painting for Sunyaev-Zel'dovich Analyses: Transfer Function vs. Hybrid Effective Field Theory**, *Phys. Rev. D* **113**, 063558 (2026), arXiv:2504.11794.
- [4] Liu, R. H., Ferraro, S., Schaan, E., *et al.*, **Measurements of the Thermal Sunyaev-Zel'dovich Effect with ACT and DESI Luminous Red Galaxies**, *Phys. Rev. D* **112**, 083561 (2025), arXiv:2502.08850.

- [5] **Liu, R. H.** and Breysse, P. C., **Coupling parsec and gigaparsec scales: primordial non-Gaussianity with multi-tracer intensity mapping**, *Phys. Rev. D* **103**, 063520 (2021), arXiv:2002.10483.
- [6] **Liu, R. H.**, Hill, R., Scott, D., Almaini, O., An, F., Gubbels, C., Hsu, L., Lin, L., Smail, I., Stach, S., **A machine learning approach for identifying the counterparts of submillimetre galaxies and applications to the GOODS-North field**, *MNRAS* **489**, 1770 (2019), arXiv:1901.09594.

Other Publications:

- [7] Gong, Y., Gallardo, P. A., Bean, R., Moore, J., Vavagiakis, E. M., ..., **Liu, R. H.**, ..., *et al.*, **Detection of the pairwise kinematic Sunyaev-Zel'dovich effect and pairwise velocity with DESI DR1 galaxies and ACT DR6 and Planck CMB data**, *Phys. Rev. D* **113**, 063538 (2026), arXiv:2511.23417.
- [8] Hadzhiyska, B., Ferraro, S., Ried Guachalla, B., Schaan, E., ..., **Liu, R. H.**, ..., *et al.*, **Evidence for large baryonic feedback at low and intermediate redshifts from kinematic Sunyaev-Zel'dovich observations with ACT and DESI photometric galaxies**, *Phys. Rev. D* **112**, 083509 (2025), arXiv:2407.07152.

Talks and Presentations

Seminars:

Lawrence Berkeley National Laboratory CMB Lunch

MODEL-INDEPENDENT MEASUREMENTS OF BARYON FEEDBACK WITH KSZ AND LENSING

Berkeley, California

Apr. 2026

Stanford Cosmology Seminar

PROBING BARYONIC MATTER IN THE UNIVERSE WITH THE SUNYAEV-ZEL'DOVICH EFFECT

Stanford, California

Mar. 2025

Lawrence Berkeley National Laboratory CMB Lunch

PROBING BARYONIC MATTER IN THE UNIVERSE WITH THE SUNYAEV-ZEL'DOVICH EFFECT

Berkeley, California

Jan. 2025

Conference and Collaboration Talks:

New Synergies in Multi-Probe Cosmology

PROBING BOUND AND UNBOUND GAS WITHIN SIMULATIONS TO UNDERSTAND THE SUNYAEV-ZEL'DOVICH EFFECTS
(POSTER)

KITP, Santa Barbara, California

Feb. 2026

CAMELS Collaboration Telecon

FAST RECONSTRUCTION OF BARYONIC SIMULATION FIELDS USING CAMELS AND MILLENNIUMTNG

Virtual

Jan. 2026

DESI Collaboration Winter Meeting

PROBING BOUND AND UNBOUND GAS WITH THE SUNYAEV-ZEL'DOVICH EFFECTS

Tucson, Arizona

Dec. 2025

Cosmology in the Adriatic

COMPARISON OF LAGRANGIAN AND EULERIAN EFT METHODS FOR BARYON PASTING ON SIMULATIONS

Split, Croatia

Jul. 2024

DESI Collaboration Summer Meeting

DESI x ACT: CROSS CORRELATION FOR DETECTION OF THE THERMAL SUNYAEV-ZEL'DOVICH EFFECT

Marseille, France

Jul. 2024

Collaboration Memberships

Dark Energy Spectroscopic Instrument (DESI), Collaboration Member

Awards and Honours

Outstanding Graduate Student Instructor Award , UC Berkeley	2025
NSERC Postgraduate Scholarship – Doctoral (PGS-D) , Natural Sciences and Engineering Research Council of Canada	2022 - 2025
Sushant Singh Rajput Memorial Fellowship , UC Berkeley	2021
NSERC Undergraduate Student Research Award , Natural Sciences and Engineering Research Council of Canada	2018, 2019
Trek Excellence Scholarship , UBC	2017 - 2020
Awarded to students in the top 5% of their faculty and year. Received during every eligible term.	
Various Other Undergraduate Scholarships , UBC	2016 - 2021

Teaching and Mentoring

Teaching:

Department of Physics, UC Berkeley	<i>Berkeley, California</i>
GRADUATE STUDENT INSTRUCTOR	2021 - Present
- Taught courses on Electromagnetism and Optics, Bayesian Statistics and Machine Learning, and Introductory Physics. - Received the Outstanding Graduate Student Instructor Award for teaching.	
Department of Physics and Astronomy, University of British Columbia	<i>Vancouver, British Columbia</i>
UNDERGRADUATE TEACHING ASSISTANT	Sept. 2018 - Apr. 2019
- PHYS 101: Energy and Waves (taught for two terms). - Received the Undergraduate Teaching Excellence Award from the UBC Science Undergraduate Society.	

Mentorship:

Linda Jin , Post-baccalaureate Researcher, UC Berkeley (now Ph.D. student, University of Washington)	2025 - Present
Machine learning emulators of baryonic suppression of the matter power spectrum using CAMELS simulations; manuscript in preparation.	

Volunteering, Service and Outreach

Astronomy On Tap East Bay	<i>Oakland, California</i>
SPEAKER	2026
UC Berkeley Astro Night	<i>Berkeley, California</i>
VOLUNTEER	2024 - 2025
UofT AstroTours	<i>Toronto, Ontario</i>
VOLUNTEER	Jun. 2019 - Aug. 2019

Skills

Programming	Python (NumPy, SciPy, JAX, PyTorch, scikit-learn), C++, R, MATLAB, Mathematica
Computing	High-performance computing (NERSC), MPI, git, Linux
Languages	English (Native Proficiency), Mandarin Chinese (Limited Working Proficiency)